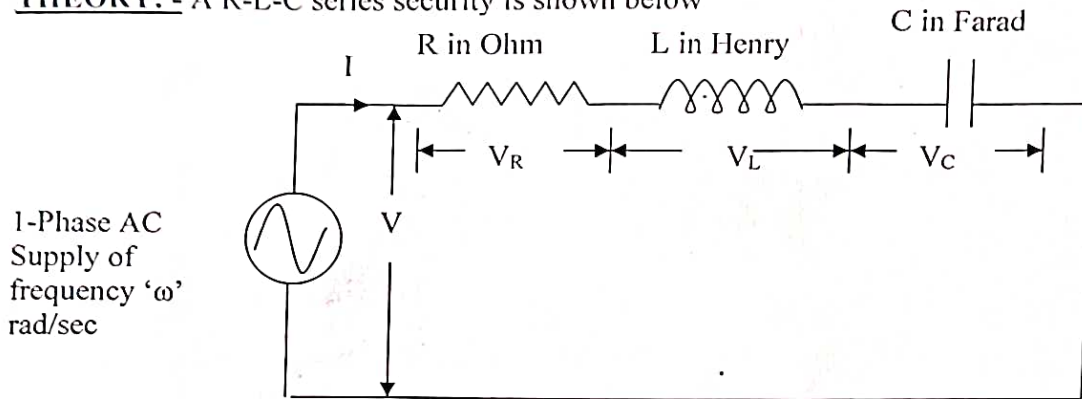


EXPERIMENT NO: - 08

AIM OF THE EXPERIMENT: - TO MEASURE CURRENT, VOLTAGE AND POWER IN A R-L-C SERIES CIRCUIT EXCITED BY SINGLE PHASE AC SUPPLY.

THEORY: - A R-L-C series security is shown below



$$\begin{aligned}
 V &= V_R + V_L + V_C = IR + I(j\omega L) + I(1/j\omega C) \\
 &= I[R + j\omega L + 1/j\omega C] = I[R + j(\omega L - 1/(\omega C))] = IZ \\
 Z &= |Z| \\
 I[R + j(\omega L - 1/(\omega C))] &= IZ \angle \theta \\
 I &= V/Z \\
 Z &= \sqrt{R^2 + (\omega L - 1/(\omega C))^2} \\
 \theta &= \tan^{-1} (\omega L - 1/(\omega C))/R \\
 \theta &= \tan^{-1} [(\omega L - 1/(\omega C))/R] \\
 I &= V/Z
 \end{aligned}$$

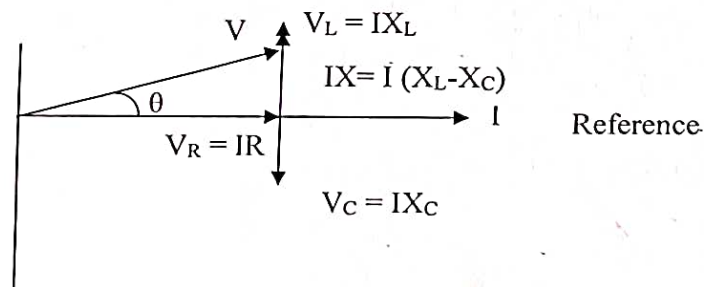
PHASOR DIAGRAM: -

(a) When $\omega L > 1/\omega C$ (Inductive circuit)

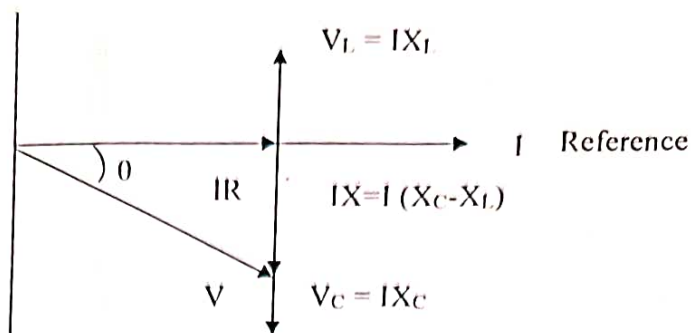
$$X_L = \omega L$$

$$X_C = 1/(\omega C)$$

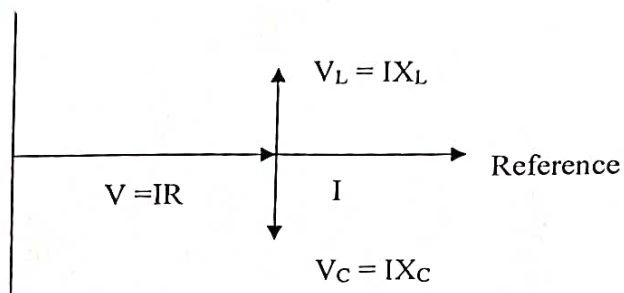
*Power factor = $\cos \theta$ lag. (Current lags supply voltage V by an angle θ .)



(b) When $\omega L < 1/\omega C$, power factor = $\cos\theta$ lead. (Current leads supply voltage V by an angle θ).
(Capacitive circuit)



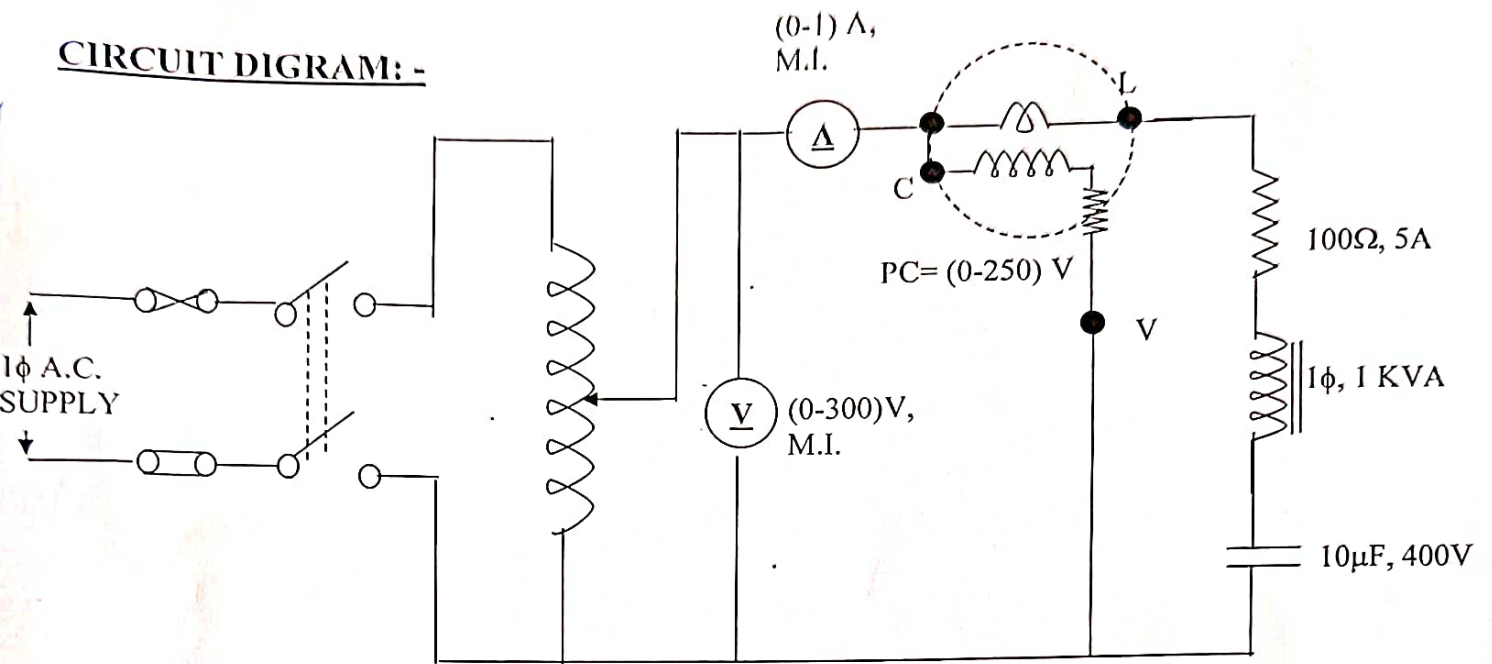
(c) When $\omega L = 1/\omega C$, The angle θ becomes zero. The circuit then behaves as a purely resistive circuit. The current through the circuit becomes in phase with the source voltage. This condition is known as resonance.
(Purely Resistive circuit)



APARATUS REQUIRED: -

SL.NO.	APPARATUS	SPECIFICATION	QUANTITY
1	VARIAC	(0-230) V, 5A	01
2	RHEOSTAT	100Ω, 2A	01
3	INDUCTOR	1-PHASE, 1KVA, 230V, 4A	01
4	CAPACITOR	10μF, 440 V	01
5	VOLTMETER	(0-230) V MI TYPE	01
6	AMMETER	(0-1) A MI TYPE	01
7	WATTMETER	(0-75) W	01
8	CONNECTING WIRE	3/22 SWG, INSULATED CU WIRE	AS REQUIRED

CIRCUIT DIGRAM: -



PROCEDURE: -

1. MAKE CONNECTIONS AS PER CIRCUIT DIAGRAM (FOR ONLY ONE CONDITION).
2. KEEP THE VARIAC IN ZERO POSITION AND SWITCH ON THE SUPPLY.
3. VARY THE VOLTAGE FROM 0 TO 230V IN SUITABLE STEPS.
4. TAKE THE CORRESPONDING READINGS OF VOLTMETER, AMMETER AND WATTMETER.
5. REPEAT STEPS 3-5 AND TAKE 6-8 READINGS. FIND TRUE POWER MULTIPLYING THE WATTMETER READING BY MULTIPLYING FACTOR THEN TABULATE.

TABULATION: -

<u>SL.NO.</u>	<u>VOLTMETER</u> <u>READING (V)</u> <u>IN VOLT.</u>	<u>AMMETER</u> <u>READING (I)</u> <u>IN AMP.</u>	<u>WATTMETER READING (W)</u> <u>IN WATT</u>	<u>CosΦ</u>	<u>Mean CosΦ</u> $= \frac{\sum_{i=1}^N (\text{Cos}\Phi)_i}{N}$
<u>1</u>					
<u>2</u>					
<u>3</u>					
<u>4</u>					
<u>5</u>					
<u>6</u>					
<u>7</u>					
<u>8</u>					

Multiplying factor = (current range used*voltage range used) / wattmeter range.

CALCULATION: -

CONCLUSION: -

PRECAUTIONS: -

1. ALL THE CONNECTIONS SHOULD BE RIGHT & TIGHT.
2. THE VOLTMETER, AMMETER AND WATTMETER SHOULD BE CAREFULLY CHOSEN SO THAT THEIR RANGES ARE MORE THAN THE MAXIMUM VALUES TO BE MEASURED.
3. SPECIAL CARE SHOULD BE TAKEN TO USE UPPER AND LOWER LIMLITS OF VOLTAGE, CURRENT, POWER.
4. SUPPLY SHOULD BE SWITCHED ON AFTER ENSURING CORRECTNESS OF CONNECTION.
5. ANY LIVE TERMINAL SHOULD NOT BE TOUCHED WHILE SUPPLY IS ON.

QUESTIONS: -

1. WHAT DO YOU UNDERSTAND BY THE TERM POWER FACTOR IN REFERENCE TO A.C CIRCUIT?
2. DEFINE THE TERM RESONANCE?
3. WHAT IS CONDUCTANCE & SUSCEPTANCE FOR AN A.C CIRCUIT?
4. WHAT IS QUALITY FACTOR OF A COIL?
5. FIND THE VALUE OF FREQUENCY AT WHICH A R-L-C SERIES CIRCUIT WILL HAVE RESONANCE?